

2) Linear Topology :-

a. Command :-

- i. `sudo su`
- ii. `mn --topo linear,3`

```
Sep 24 11:55
root@sagnik-bachhar-VirtualBox: /home/sagnik-bachhar — sudo su

sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ sudo su
[sudo: authenticate] Password:
root@sagnik-bachhar-VirtualBox:/home/sagnik-bachhar# mn --topo linear,3
*** No default OpenFlow controller found for default switch!
*** Falling back to OVS Bridge
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2 h3
*** Adding switches:
s1 s2 s3
*** Adding links:
(h1, s1) (h2, s2) (h3, s3) (s2, s1) (s3, s2)
*** Configuring hosts
h1 h2 h3
*** Starting controller

*** Starting 3 switches
s1 s2 s3 ...
*** Starting CLI:
mininet> nodes
available nodes are:
h1 h2 h3 s1 s2 s3
mininet> links
h1-eth0<->s1-eth1 (OK OK)
h2-eth0<->s2-eth1 (OK OK)
h3-eth0<->s3-eth1 (OK OK)
s2-eth2<->s1-eth2 (OK OK)
s3-eth2<->s2-eth3 (OK OK)
mininet> net
h1 h1-eth0:s1-eth1
```

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[sudo: authenticate] Password:
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h1 h2 h3
*** Adding switches:
s1 s2 s3
*** Adding links:
(h1, s1) (h2, s2) (h3, s3) (s2, s1) (s3, s2)
*** Configuring hosts
h1 h2 h3
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s1 s2 s3 ...
*** Starting CLI:
mininet> nodes
available nodes are:
h1 h2 h3 s1 s2 s3
mininet> links
h1-eth0<->s1-eth1 (OK OK)
h2-eth0<->s2-eth1 (OK OK)
h3-eth0<->s3-eth1 (OK OK)
s2-eth2<->s1-eth2 (OK OK)
s3-eth2<->s2-eth3 (OK OK)
mininet> net
h1 h1-eth0:s1-eth1
h2 h2-eth0:s2-eth1
h3 h3-eth0:s3-eth1
s1 lo: s1-eth1:h1-eth0 s1-eth2:s2-eth2
s2 lo: s2-eth1:h2-eth0 s2-eth2:s1-eth2 s2-eth3:s3-eth2
s3 lo: s3-eth1:h3-eth0 s3-eth2:s2-eth3
mininet> dump
<Host h1: h1-eth0:10.0.0.1 pid=6920>
<Host h2: h2-eth0:10.0.0.2 pid=6922>
<Host h3: h3-eth0:10.0.0.3 pid=6924>
<OVSBridge s1: lo:127.0.0.1,s1-eth1:None,s1-eth2:None pid=6929>
<OVSBridge s2: lo:127.0.0.1,s2-eth1:None,s2-eth2:None,s2-eth3:None pid=6932>
<OVSBridge s3: lo:127.0.0.1,s3-eth1:None,s3-eth2:None pid=6935>
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3
h2 -> h1 h3
h3 -> h1 h2
*** Results: 0% dropped (6/6 received)
mininet> h1 ping -c 4 h3
PING 10.0.0.3 (10.0.0.3) 56(84) bytes of data.
64 bytes from 10.0.0.3: icmp_seq=1 ttl=64 time=0.227 ms
64 bytes from 10.0.0.3: icmp_seq=2 ttl=64 time=0.046 ms
64 bytes from 10.0.0.3: icmp_seq=3 ttl=64 time=0.060 ms
64 bytes from 10.0.0.3: icmp_seq=4 ttl=64 time=0.065 ms

--- 10.0.0.3 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3090ms
```

```
Sep 24 11:56
root@sagnik-bachhar-VirtualBox: /home/sagnik-bachhar — sudo su

sagnik-bachhar@sagnik-bachhar:~$ mininet> ping 10.0.0.3
--- 10.0.0.3 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3090ms
rtt min/avg/max/mdev = 0.046/0.099/0.227/0.073 ms
mininet> exit
*** Stopping 0 controllers

*** Stopping 5 links
.^C

Keyboard Interrupt. Shutting down and cleaning up...

*** Removing excess controllers/ofprotocols/ofdatapaths/pings/noxes
killall controller ofprotocol ofdatapath ping nox_corelt-nox_core ovs-openflowd ovs-controllerovs-testcontroller udpbwte
st mnexec ivs ryu-manager 2> /dev/null
killall -9 controller ofprotocol ofdatapath ping nox_corelt-nox_core ovs-openflowd ovs-controllerovs-testcontroller udpb
wtest mnexec ivs ryu-manager 2> /dev/null
pkill -9 -f "sudo mnexec"
*** Removing junk from /tmp
rm -f /tmp/vconn* /tmp/vlogs* /tmp/*.out /tmp/*.log
*** Removing old X11 tunnels
*** Removing excess kernel datapaths
ps ax | egrep -o 'dp[0-9]+' | sed 's/dp/nl:/'
*** Removing OVS datapaths
ovs-vsctl --timeout=1 list-br
ovs-vsctl --timeout=1 list-br
*** Removing all links of the pattern foo-ethX
ip link show | egrep -o '([_[:alnum:]]+-eth[[:digit:]]+)'
ip link show
*** Killing stale mininet node processes
pkill -9 -f mininet;
```

```
Sep 24 11:56
root@sagnik-bachhar-VirtualBox: /home/sagnik-bachhar — sudo su

sagnik-bachhar@sagnik-bachhar:~$ mininet> ping 10.0.0.3
--- 10.0.0.3 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3090ms
rtt min/avg/max/mdev = 0.046/0.099/0.227/0.073 ms
mininet> exit
*** Stopping 0 controllers

*** Stopping 5 links
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Keyboard Interrupt. Shutting down and cleaning up...

*** Removing excess controllers/ofprotocols/ofdatapaths/pings/noxes
killall controller ofprotocol ofdatapath ping nox_corelt-nox_core ovs-openflowd ovs-controllerovs-testcontroller udpbwte
st mnexec ivs ryu-manager 2> /dev/null
killall -9 controller ofprotocol ofdatapath ping nox_corelt-nox_core ovs-openflowd ovs-controllerovs-testcontroller udpb
wtest mnexec ivs ryu-manager 2> /dev/null
pkill -9 -f "sudo mnexec"
*** Removing junk from /tmp
rm -f /tmp/vconn* /tmp/vlogs* /tmp/*.out /tmp/*.log
*** Removing old X11 tunnels
*** Removing excess kernel datapaths
ps ax | egrep -o 'dp[0-9]+' | sed 's/dp/nl:/'
*** Removing OVS datapaths
ovs-vsctl --timeout=1 list-br
ovs-vsctl --timeout=1 list-br
*** Removing all links of the pattern foo-ethX
ip link show | egrep -o '([_[:alnum:]]+-eth[[:digit:]]+)'
ip link show
*** Killing stale mininet node processes
pkill -9 -f mininet:
*** Shutting down stale tunnels
pkill -9 -f Tunnel=Ethernet
pkill -9 -f .ssh/mn
rm -f ~/.ssh/mn/*
*** Cleanup complete.
root@sagnik-bachhar-VirtualBox: /home/sagnik-bachhar#
```